Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Castelanelli Brothers Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
401 W. Armstrong RD	Lodi	San Joaquin	95242

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/25/1928

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X060-XX32-XXXX X058-X060-XX35-XXXX

B. OPERATORS

Castelanelli, Larry

Operator name: Castelanelli, Larry

Telephone no.: Landline: ; Cellular: (209) 747-7727

Mailing Address Number and Street	City	State	Zip Code
401 Armstrong RD	Lodi	CA	95242

This operator is responsible for paying permit fees.

C. OWNERS

Castelanelli, Larry

Legal owner name: Castelanelli, Larry

Telephone no.: Landline: ; Cellular: (209) 747-7727

Mailing Address Number and Street	City	State	Zip Code
401 Armstrong RD	Lodi	CA	95242

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	0	0	0
Average number	0	0	0	0	0	0
Avg live weight (lbs)	0	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1.00 *tons per reporting period*

Total nitrogen from manure: 1.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 0.70 *lbs per reporting period*

Total phosphorus from manure: 1.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW 1	Ground water
IW 10	Ground water
IW 11	Ground water
IW 2	Ground water
IW 3	Ground water

Source Description	Type
IW 4	Ground water
IW 5	Ground water
IW 6	Ground water
IW 7	Ground water
IW 9	Ground water
Woodbridge Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/09/2023	Solid commercial fertilizer *UN32*	58.45 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	37,370.59	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	37,370.59	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	52	52	0	none	X058-X060-XX52-XXXX X058-X060-XX55-XXXX
2	35	35	2	none	X058-X060-XX25-XXXX X058-X060-XX26-XXXX
3N	60	60	1	none	X058-X060-XX31-XXXX X058-X060-XX32-XXXX
3S	52	52	2	none	X058-X060-XX31-XXXX X058-X060-XX32-XXXX
4	60	60	0	none	X058-X060-XX49-XXXX X058-X060-XX50-XXXX X058-X060-XX51-XXXX
5	40	40	0	none	X058-X060-XX35-XXXX X058-X060-XX44-XXXX X058-X090-XX18-XXXX X058-X090-XX21-XXXX
7	40	40	2	none	X058-X060-XX12-XXXX
8	80	80	2	none	X055-X230-XX15-XXXX
9	80	80	1	none	X058-X060-XX34-XXXX X058-X090-XX05-XXXX
Totals for areas that were used for application					
Totals for areas that were not used for application	499	499	10		
Land application area totals	499	499	10		

B. CROPS AND HARVESTS

2

Field name: 2

2

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	469.00 *ton*	As-is		68.4	5,720.00	1,030.00	5,740.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	13.40	153.30	27.60	153.83	626.69

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,050.00 *ton*	As-is		69.4	3,860.00	440.00	3,730.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	231.60	26.40	223.80	1,119.96

3N

Field name: 3N

10/25/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 60 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	432.00 *ton*	As-is		19.8	26,100.00	2,450.00	33,900.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.20	375.84	35.28	488.16	1,570.64

3S

Field name: 3S

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	252.00 *ton*	As-is		66.2	5,200.00	1,170.00	6,830.00		8.07

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	12.60	131.04	29.48	172.12	687.37

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	590.00 *ton*	As-is		68.0	3,680.00	520.00	3,810.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.50	217.12	30.68	224.79	962.88

7

Field name: 7

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	564.00 *ton*	As-is		69.0	4,680.00	980.00	5,670.00		8.81

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	14.10	131.98	27.64	159.89	770.17

7

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,220.00 *ton*	As-is		71.4	3,690.00	440.00	3,640.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.50	225.09	26.84	222.04	1,849.28

8

Field name: 8

10/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	1,040.00 *ton*	As-is		66.4	5,030.00	1,070.00	6,200.00		7.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	13.00	130.78	27.82	161.20	683.16

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	2,416.00 *ton*	As-is		69.4	3,920.00	440.00	3,720.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.20	236.77	26.58	224.69	924.12

9

Field name: 9

10/27/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 80 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	560.00 *ton*	As-is		20.4	25,000.00	2,860.00	31,800.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.00	350.00	40.04	445.20	1,359.57

NUTRIENT BUDGET

A. LAND APPLICATIONS

2 - 10/18/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	11.67	0.00	0.00	154.50	3,600,000.00 *gal*
Application event totals		11.67	0.00	0.00	154.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	3.48	0.00	0.00	143.77	1,800,000.00 *gal*
Application event totals		3.48	0.00	0.00	143.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	70.40	0.00	0.00	0.00	
Application event totals		70.40	0.00	0.00	0.00	

2 - 05/17/2023: Corn, silage

Field name: 2

Crop: Corn, silage Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.84	0.00	0.00	365.42	4,575,000.00 *gal*
Application event totals		8.84	0.00	0.00	365.42	

2 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.40	0.00	0.00	347.45	4,350,000.00 *gal*
Application event totals		8.40	0.00	0.00	347.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	102.40	0.00	0.00	0.00	
Application event totals		102.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.11	0.00	0.00	335.47	4,200,000.00 *gal*
Application event totals		8.11	0.00	0.00	335.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.68	0.00	0.00	317.50	3,975,000.00 *gal*
Application event totals		7.68	0.00	0.00	317.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.82	0.00	0.00	323.49	4,050,000.00 *gal*
Application event totals		7.82	0.00	0.00	323.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.68	0.00	0.00	317.50	3,975,000.00 *gal*
Application event totals		7.68	0.00	0.00	317.50	

2 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.53	0.00	0.00	311.51	3,900,000.00 *gal*
Application event totals		7.53	0.00	0.00	311.51	

3N - 10/25/2022: Alfalfa, hay

Field name: 3N

Crop: Alfalfa, hay

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.61	0.00	0.00	253.35	6,624,000.00 *gal*
Application event totals		3.61	0.00	0.00	253.35	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.56	0.00	0.00	249.68	6,528,000.00 *gal*
Application event totals		3.56	0.00	0.00	249.68	

05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	51.20	0.00	0.00	0.00	
Application event totals		51.20	0.00	0.00	0.00	

05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.45	0.00	0.00	242.34	6,336,000.00 *gal*
Application event totals		3.45	0.00	0.00	242.34	

3N - 10/25/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.51	0.00	0.00	246.01	6,432,000.00 *gal*
Application event totals		3.51	0.00	0.00	246.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.77	0.00	0.00	264.37	6,912,000.00 *gal*
Application event totals		3.77	0.00	0.00	264.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.66	0.00	0.00	257.03	6,720,000.00 *gal*
Application event totals		3.66	0.00	0.00	257.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.61	0.00	0.00	253.35	6,624,000.00 *gal*
Application event totals		3.61	0.00	0.00	253.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.51	0.00	0.00	246.01	6,432,000.00 *gal*
Application event totals		3.51	0.00	0.00	246.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.40	0.00	0.00	238.67	6,240,000.00 *gal*
Application event totals		3.40	0.00	0.00	238.67	

3S - 10/18/2022: Oats, silage-soft dough

3S - 10/18/2022: Oats, silage-soft dough

Field name: 3S

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	10.21	0.00	0.00	135.19	1,800,000.00 *gal*
Application event totals		10.21	0.00	0.00	135.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	5.07	0.00	0.00	209.67	1,500,000.00 *gal*
Application event totals		5.07	0.00	0.00	209.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	70.40	0.00	0.00	0.00	
Application event totals		70.40	0.00	0.00	0.00	

3S - 05/17/2023: Corn, silage

Field name: 3S

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.11	0.00	0.00	335.47	2,400,000.00 *gal*
Application event totals		8.11	0.00	0.00	335.47	

3S - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.11	0.00	0.00	335.47	2,400,000.00 *gal*
Application event totals		8.11	0.00	0.00	335.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	102.40	0.00	0.00	0.00	
Application event totals		102.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.87	0.00	0.00	366.92	2,625,000.00 *gal*
Application event totals		8.87	0.00	0.00	366.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.36	0.00	0.00	345.95	2,475,000.00 *gal*
Application event totals		8.36	0.00	0.00	345.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.11	0.00	0.00	335.47	2,400,000.00 *gal*
Application event totals		8.11	0.00	0.00	335.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.60	0.00	0.00	314.50	2,250,000.00 *gal*
Application event totals		7.60	0.00	0.00	314.50	

3S - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.35	0.00	0.00	304.02	2,175,000.00 *gal*
Application event totals		7.35	0.00	0.00	304.02	

7 - 10/18/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	7.72	0.00	0.00	0.00	4,048,200.00 *gal*
Application event totals		7.72	0.00	0.00	0.00	

02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	0.52	0.00	0.00	204.64	2,179,800.00 *gal*
Application event totals		0.52	0.00	0.00	204.64	

02/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	70.40	0.00	0.00	0.00	
Application event totals		70.40	0.00	0.00	0.00	

7 - 05/17/2023: Corn, silage

Field name: 7

Crop: Corn, silage Plant date: 05/17/2023

7 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.22	0.00	0.00	477.50	5,086,200.00 *gal*
Application event totals		1.22	0.00	0.00	477.50	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.17	0.00	0.00	458.01	4,878,600.00 *gal*
Application event totals		1.17	0.00	0.00	458.01	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	102.40	0.00	0.00	0.00	
Application event totals		102.40	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.16	0.00	0.00	453.14	4,826,700.00 *gal*
Application event totals		1.16	0.00	0.00	453.14	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.17	0.00	0.00	458.01	4,878,600.00 *gal*
Application event totals		1.17	0.00	0.00	458.01	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.15	0.00	0.00	448.26	4,774,800.00 *gal*
Application event totals		1.15	0.00	0.00	448.26	

7 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.12	0.00	0.00	438.52	4,671,000.00 *gal*
Application event totals		1.12	0.00	0.00	438.52	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.10	0.00	0.00	428.77	4,567,200.00 *gal*
Application event totals		1.10	0.00	0.00	428.77	

8 - 10/19/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	9.15	0.00	0.00	121.11	6,450,000.00 *gal*
Application event totals		9.15	0.00	0.00	121.11	

02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	4.56	0.00	0.00	188.70	5,400,000.00 *gal*
Application event totals		4.56	0.00	0.00	188.70	

02/17/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	70.40	0.00	0.00	0.00	
Application event totals		70.40	0.00	0.00	0.00	

8 - 05/18/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.24	0.00	0.00	340.71	9,750,000.00 *gal*
Application event totals		8.24	0.00	0.00	340.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.36	0.00	0.00	345.95	9,900,000.00 *gal*
Application event totals		8.36	0.00	0.00	345.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	102.40	0.00	0.00	0.00	
Application event totals		102.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.30	0.00	0.00	343.33	9,825,000.00 *gal*
Application event totals		8.30	0.00	0.00	343.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.24	0.00	0.00	340.71	9,750,000.00 *gal*
Application event totals		8.24	0.00	0.00	340.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	8.05	0.00	0.00	332.85	9,525,000.00 *gal*
Application event totals		8.05	0.00	0.00	332.85	

8 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.92	0.00	0.00	327.61	9,375,000.00 *gal*
Application event totals		7.92	0.00	0.00	327.61	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	7.60	0.00	0.00	314.50	9,000,000.00 *gal*
Application event totals		7.60	0.00	0.00	314.50	

9 - 10/27/2022: Alfalfa, hay

Field name: 9

Crop: Alfalfa, hay

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.86	0.00	0.00	316.00	8,910,000.00 *gal*
Application event totals		5.86	0.00	0.00	316.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.68	0.00	0.00	306.43	8,640,000.00 *gal*
Application event totals		5.68	0.00	0.00	306.43	

05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	51.20	0.00	0.00	0.00	
Application event totals		51.20	0.00	0.00	0.00	

9 - 10/27/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.56	0.00	0.00	300.04	8,460,000.00 *gal*
Application event totals		5.56	0.00	0.00	300.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.62	0.00	0.00	303.24	8,550,000.00 *gal*
Application event totals		5.62	0.00	0.00	303.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.91	0.00	0.00	319.20	9,000,000.00 *gal*
Application event totals		5.91	0.00	0.00	319.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	3.72	0.00	0.00	260.76	9,090,000.00 *gal*
Application event totals		3.72	0.00	0.00	260.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	1.38	0.00	0.00	292.77	8,910,000.00 *gal*
Application event totals		1.38	0.00	0.00	292.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	1.35	0.00	0.00	286.85	8,730,000.00 *gal*
Application event totals		1.35	0.00	0.00	286.85	

9 - 10/27/2022: Alfalfa, hay

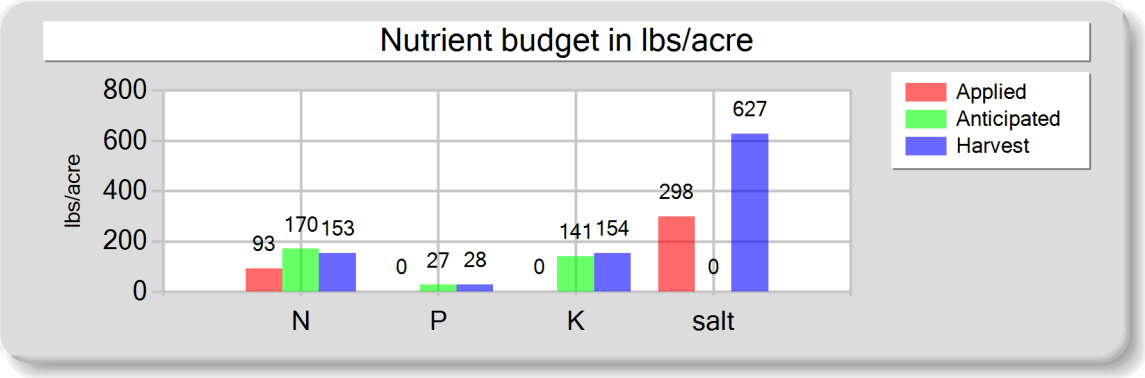
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	1.33	0.00	0.00	283.90	8,640,000.00 *gal*
Application event totals		1.33	0.00	0.00	283.90	

B. NUTRIENT BUDGET

2 - 10/18/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.15	0.00	0.00	298.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.55	0.00	0.00	298.27
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	153.30	27.60	153.83	626.69
Nutrient balance	-60.75	-27.60	-153.83	-328.42
Applied to removed ratio	0.60	0.00	0.00	0.48

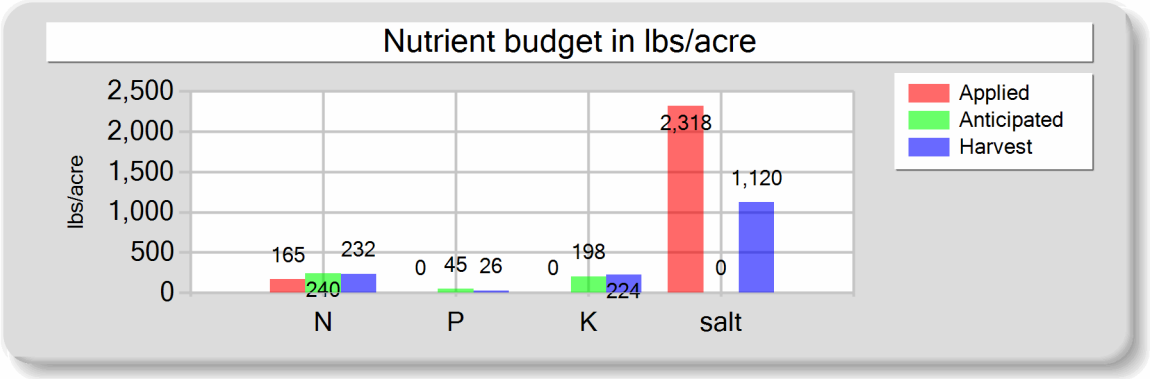
Fresh water applied	
5,400,000.00	*gallons*
198.86	*acre-inches*
5.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/17/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	102.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	56.06	0.00	0.00	2,318.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.46	0.00	0.00	2,318.33
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.60	26.40	223.80	1,119.96
Nutrient balance	-66.14	-26.40	-223.80	1,198.37
Applied to removed ratio	0.71	0.00	0.00	2.07

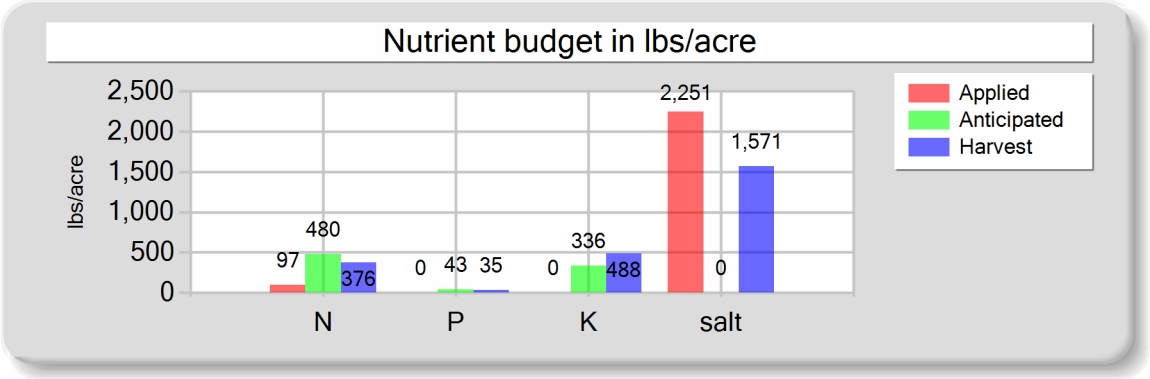
Fresh water applied	
29,025,000.00	*gallons*
1,068.89	*acre-inches*
30.54	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3N - 10/25/2022: Alfalfa, hay

Field name: 3N | Crop: Alfalfa, hay | Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	32.08	0.00	0.00	2,250.81
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	97.28	0.00	0.00	2,250.81
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	375.84	35.28	488.16	1,570.64
Nutrient balance	-278.56	-35.28	-488.16	680.18
Applied to removed ratio	0.26	0.00	0.00	1.43

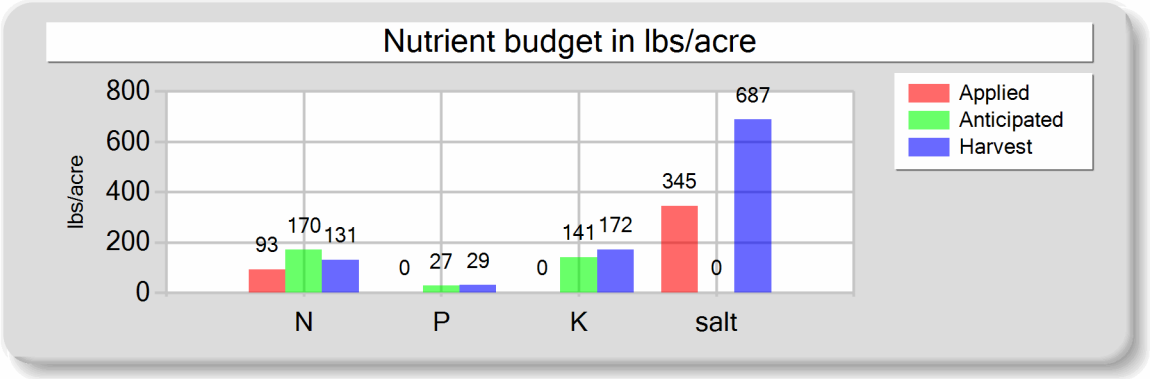
Fresh water applied	
58,848,000.00	*gallons*
2,167.17	*acre-inches*
36.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3S - 10/18/2022: Oats, silage-soft dough

Field name: 3S | Crop: Oats, silage-soft dough | Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.28	0.00	0.00	344.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.68	0.00	0.00	344.86
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	131.04	29.48	172.12	687.37
Nutrient balance	-38.36	-29.48	-172.12	-342.51
Applied to removed ratio	0.71	0.00	0.00	0.50

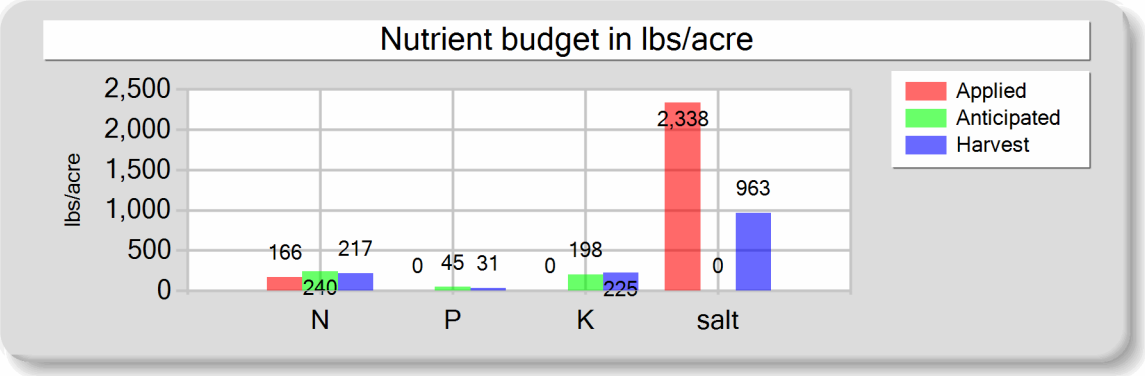
Fresh water applied	
3,300,000.00	*gallons*
121.53	*acre-inches*
6.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3S - 05/17/2023: Corn, silage

Field name: 3S | Crop: Corn, silage | Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	102.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	56.53	0.00	0.00	2,337.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.93	0.00	0.00	2,337.80
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.12	30.68	224.79	962.88
Nutrient balance	-51.19	-30.68	-224.79	1,374.92
Applied to removed ratio	0.76	0.00	0.00	2.43

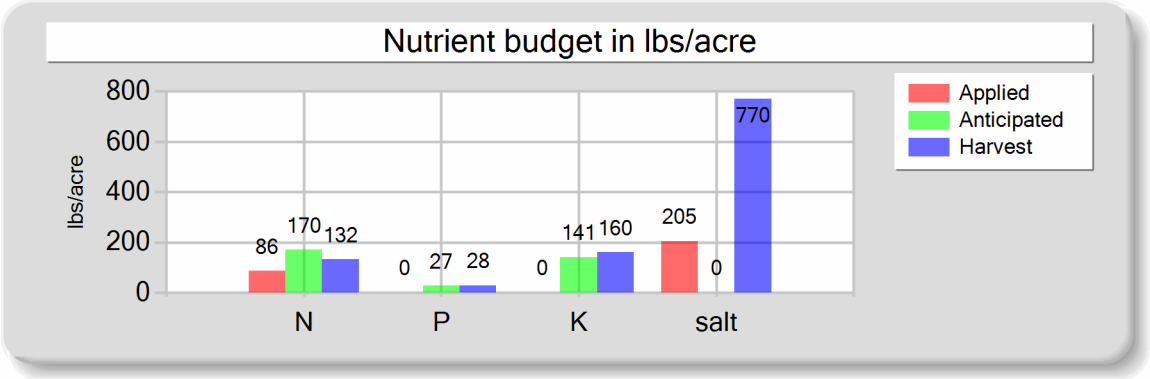
Fresh water applied	
16,725,000.00	*gallons*
615.92	*acre-inches*
30.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 10/18/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.24	0.00	0.00	204.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	85.64	0.00	0.00	204.64
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	131.98	27.64	159.89	770.17
Nutrient balance	-46.33	-27.64	-159.89	-565.53
Applied to removed ratio	0.65	0.00	0.00	0.27

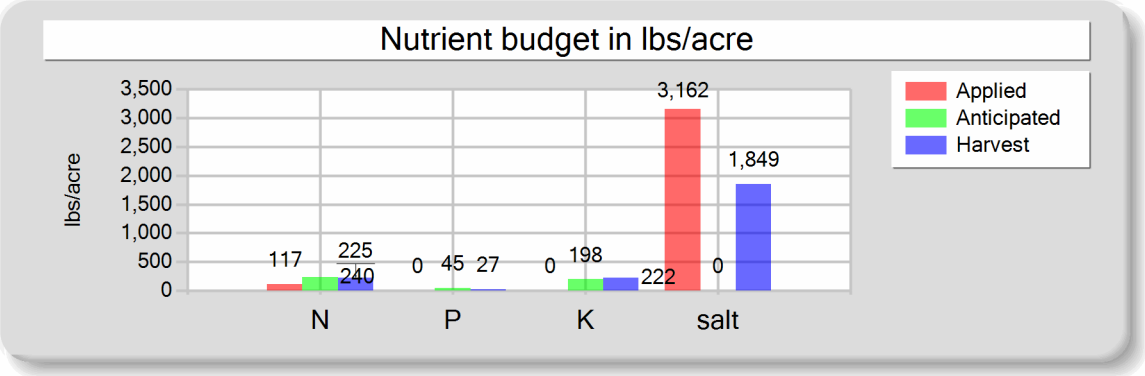
Fresh water applied	
6,228,000.00	*gallons*
229.36	*acre-inches*
5.73	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/17/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	102.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.08	0.00	0.00	3,162.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	117.48	0.00	0.00	3,162.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.09	26.84	222.04	1,849.28
Nutrient balance	-107.61	-26.84	-222.04	1,312.94
Applied to removed ratio	0.52	0.00	0.00	1.71

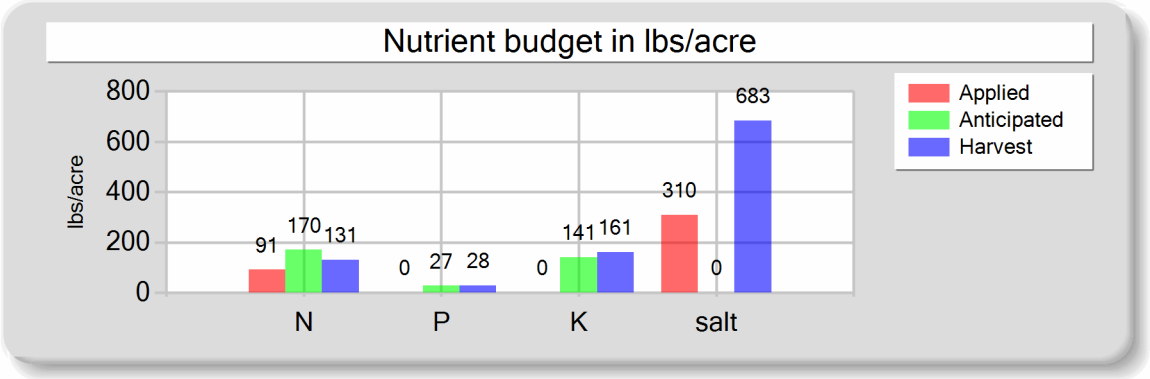
Fresh water applied	
33,683,100.00	*gallons*
1,240.43	*acre-inches*
31.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 10/19/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.71	0.00	0.00	309.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	91.11	0.00	0.00	309.81
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	130.78	27.82	161.20	683.16
Nutrient balance	-39.67	-27.82	-161.20	-373.35
Applied to removed ratio	0.70	0.00	0.00	0.45

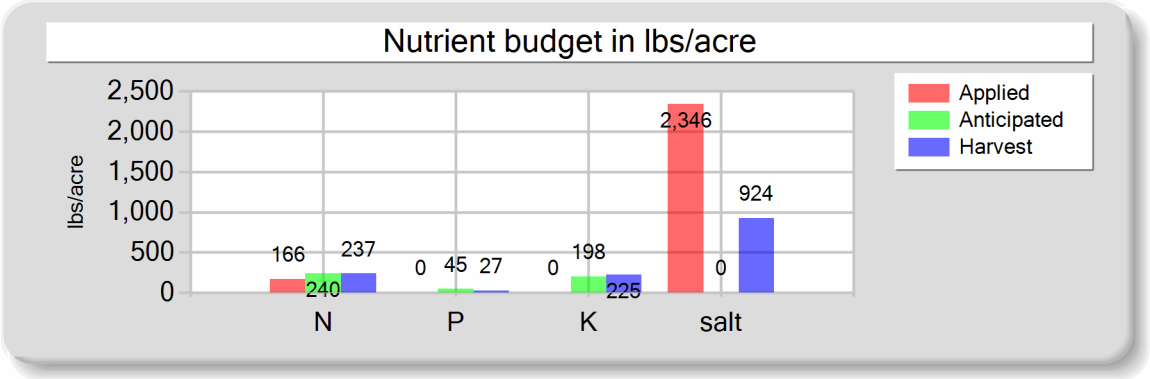
Fresh water applied	
11,850,000.00	*gallons*
436.40	*acre-inches*
5.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 05/18/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	102.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	56.72	0.00	0.00	2,345.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	166.12	0.00	0.00	2,345.66
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.77	26.58	224.69	924.12
Nutrient balance	-70.65	-26.58	-224.69	1,421.54
Applied to removed ratio	0.70	0.00	0.00	2.54

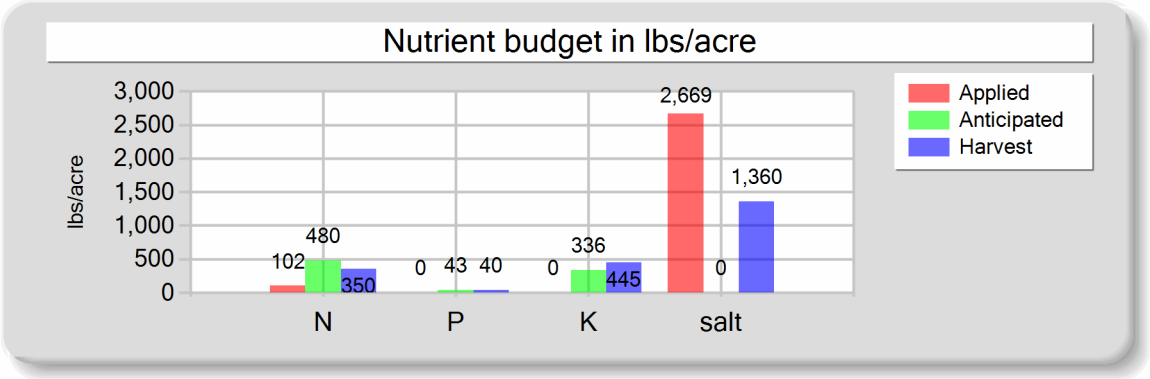
Fresh water applied	
67,125,000.00	*gallons*
2,471.99	*acre-inches*
30.90	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/27/2022: Alfalfa, hay

Field name: 9 Crop: Alfalfa, hay Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	51.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	36.40	0.00	0.00	2,669.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	101.60	0.00	0.00	2,669.18
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	350.00	40.04	445.20	1,359.57
Nutrient balance	-248.40	-40.04	-445.20	1,309.62
Applied to removed ratio	0.29	0.00	0.00	1.96

Fresh water applied	
78,930,000.00	*gallons*
2,906.72	*acre-inches*
36.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

IW 1

IW 1

Sample description: IW 1

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.70	0.00	0.00								872.00	530
DL	0.50	0.50	0.50								10.00	10

IW 10

IW 10

Sample description: IW 10

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.80	0.00	0.00								600.00	375
DL	0.50	0.50	0.50								10.00	10

IW 11

IW 11

IW 11

Sample description: IW 11

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.37	0.00	0.00								571.00	345
DL	0.50	0.50	0.50								10.00	10

IW 2

IW 2

Sample description: IW 2

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.15	0.00	0.00								702.00	450
DL	0.50	0.50	0.50								10.00	10

IW 3

IW 3

Sample description: IW 3

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.50	0.00	0.00								484.00	290
DL	0.50	0.50	0.50								10.00	10

IW 4

IW 4

IW 4

Sample description: IW 4

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.10	0.00	0.00								566.00	335
DL	0.50	0.50	0.50								10.00	10

IW 5

IW 5

Sample description: IW 5

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.92	0.00	0.00								417.00	275
DL	0.50	0.50	0.50								10.00	10

IW 6

IW 6

Sample description: IW 6

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.20	0.00	0.00								687.00	415
DL	0.50	0.50	0.50								10.00	10

IW 7

IW 7

IW 7

Sample description: IW 7

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.30	0.00	0.00								570.00	340
DL	0.50	0.50	0.50								10.00	10

IW 9

IW 9

Sample description: IW 9

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.48	0.00	0.00								518.00	315
DL	0.50	0.50	0.50								10.00	10

Woodbridge Canal

Woodbridge Canal

Sample description: Woodbridge Canal

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.60	53
DL	1.00	1.00	1.00	0.50	0.50	0.50	0.50	0.50	0.50	0.50	10.00	10

D. SOIL ANALYSES

2

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			75.00				
DL			4.00				

3N

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			80.00				
DL			4.00				

3S

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			119.00				
DL			4.00				

7

7

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			110.00				
DL			4.00				

8

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			4.00				

9

Soils

Sample and source description: Soils

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

2 - 10/18/2022: Oats, silage-soft dough

2 - 10/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,720.00	1,030.00	5,740.00		7.40
DL	100.00	100.00	100.00		0.01

2 - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,860.00	440.00	3,730.00		6.10
DL	100.00	100.00	100.00		0.01

3N - 10/25/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 19.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,100.00	2,450.00	33,900.00		13.60
DL	100.00	100.00	100.00		0.01

3S - 10/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,170.00	6,830.00		8.07
DL	100.00	100.00	100.00		0.01

3S - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,680.00	520.00	3,810.00		5.10
DL	100.00	100.00	100.00		0.01

7 - 10/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,680.00	980.00	5,670.00		8.81
DL	100.00	100.00	100.00		0.01

7 - 05/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,690.00	440.00	3,640.00		10.60
DL	100.00	100.00	100.00		0.01

8 - 05/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	440.00	3,720.00		5.00
DL	100.00	100.00	100.00		0.01

9 - 10/27/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 20.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	2,860.00	31,800.00		12.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

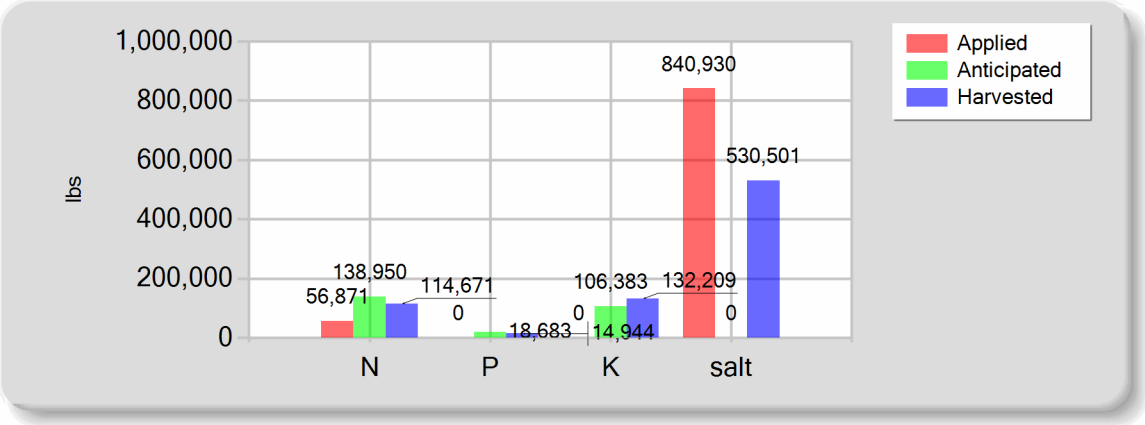
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

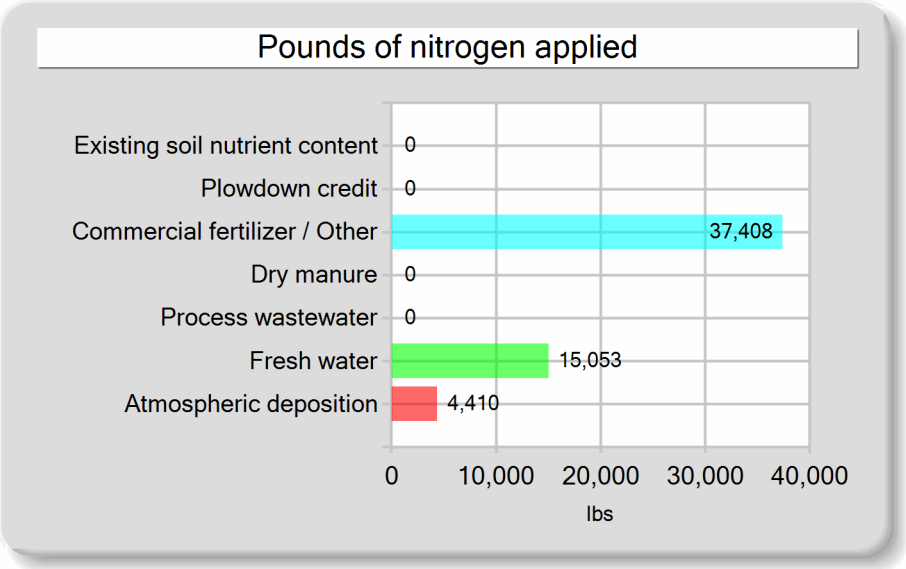
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

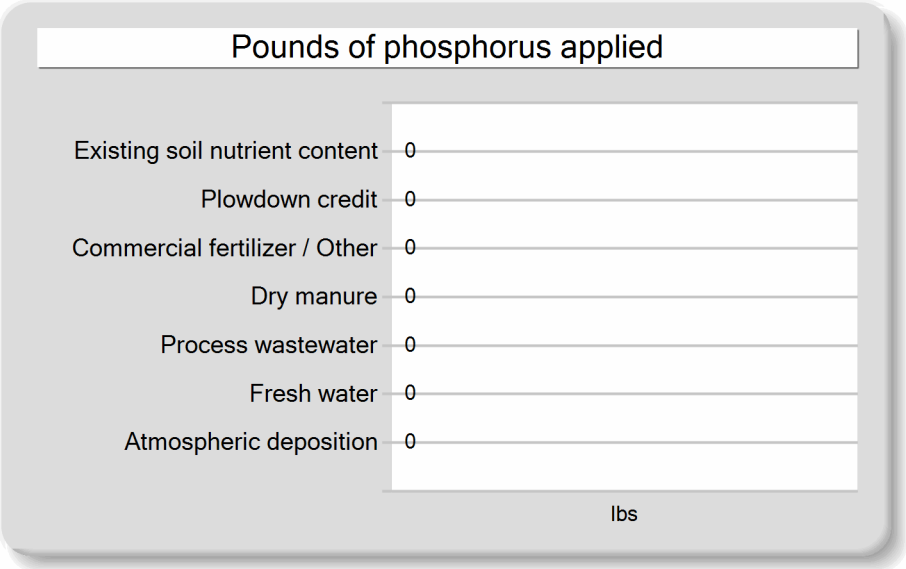
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	37,408.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15,052.71	0.00	0.00	840,929.66
Atmospheric deposition	4,410.00	0.00	0.00	0.00
Total nutrients applied	56,870.71	0.00	0.00	840,929.66
Anticipated crop nutrient removal	138,950.00	18,683.00	106,382.50	0.00
Actual crop nutrient removal	114,671.44	14,944.14	132,209.24	530,501.31
Nutrient balance	-57,800.73	-14,944.14	-132,209.24	310,428.35
Applied to removed ratio	0.50	0.00	0.00	1.59

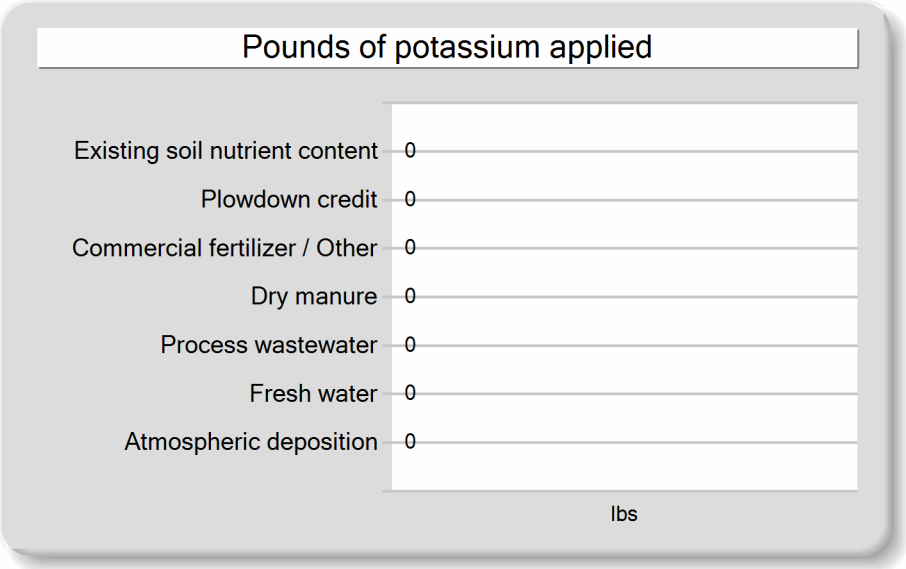
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

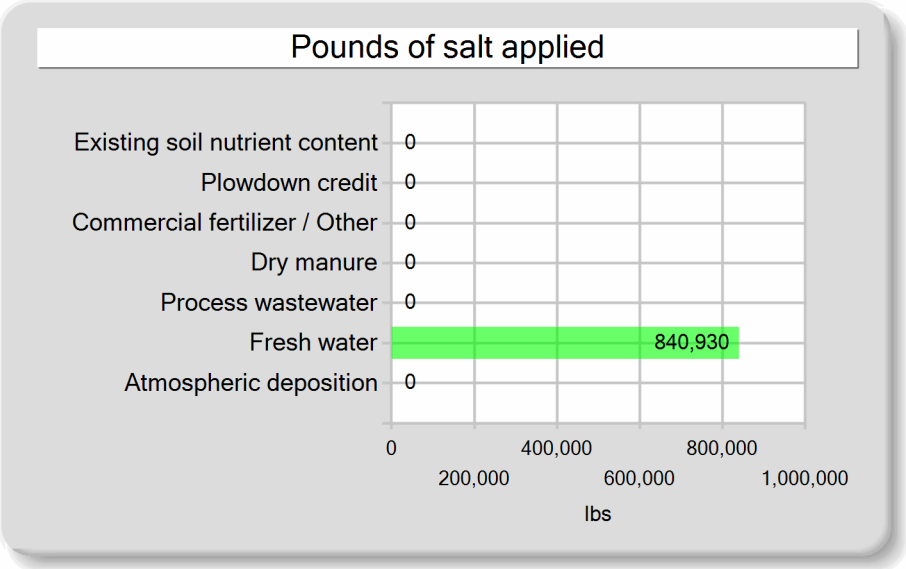


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

All animals were removed from the facility in 2020. No solid manure or wastewater remained on the facility in 2023 therefore no wastewater or solid manure samples were taken in 2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Larry Castelanelli	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Larry Castelanelli	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
2/2/24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No — Bill To: Acct # 0 — Cons # 13

Results Need By

No. Samples: — No of Bottles:

Name: Costalanelli Bros
Address: 401 W Armstrong Rd
City: Lodi State: CA Zip: 95242
Telephone: 209 747-7727 Fax:
Cell/Email:
COPY TO: F&R Ag Services
& J Sawyer @ Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCW1

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IV 1				
2	2				
3	3				
4	4				
5	5				
6	6				
7	7				
8	9				
9	10				
10	11				

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE LABORATORY INC

Report of Dairy Canal Water Analysis

Lab No.: 23I0661
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 09/08/23
Reported Date: 09/15/23
Project:
Crop ID:

Castelanelli Bros
401 W Armstrong Rd
Lodi CA 95242
00-0026187 13

E-mail: jswayer@fragservices.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	TDS mg/L	TKN mg/L	Total Nitrogen TN mg/L
1	IW 1	9/7/23	8:00	872	0.87	10.7	530	ND	10.7
2	IW 2	9/7/23	8:00	702	0.70	1.1	450	ND	1.15
3	IW 3	9/7/23	8:00	484	0.48	4.9	290	ND	5.5
4	IW 4	9/7/23	8:00	566	0.57	1.3	335	6.84	8.1
5	IW 5	9/7/23	8:00	417	0.42	3.9	275	ND	3.92
6	IW 6	9/7/23	8:00	687	0.69	8.2	415	ND	8.2
7	IW 7	9/7/23	8:00	570	0.57	6.3	340	ND	6.3
8	IW 9	9/7/23	8:00	518	0.52	1.4	315	ND	1.48
9	IW 10	9/7/23	8:00	600	0.60	6.8	375	ND	6.8
10	IW 11	9/7/23	8:00	571	0.57	6.4	345	ND	6.37

ND = None Detected

Approved By: Zoe M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: Acct # 0 Cons # 13

Results Need By

Name: Costelanelli Bros
Address: 401 W Armstrong Rd
City: Lodi State: CA Zip: 95242
Telephone: 209 747-7737 Fax:
Cell/Email:
COPY TO: F&R Ag Services
& JSawyer@Frogservices.com
REQUESTED BY: Garrett Sawyer
PROJECT:
CROP:

No. Samples: No of Bottles:

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DWW1

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW Back up				
2	Cousins				
3	Milk Barn				
4	Millers				
5	Uncles				
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Garrett Sawyer	F&R Ag Services		
Second	"	"		
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
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Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23I0665
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 09/08/23
Reported Date: 09/18/23
Project:
Crop ID:

Castelanelli Bros
401 W Armstrong Rd
Lodi CA 95242
00-0026187 13

E-mail: jswayer@fragservices.com
Copy To:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW Back Up	9/7/23	7:00	584	0.58	6.9	*		8
2	DW Cousins	9/7/23	7:00	663	0.66	1.1	*		7.9
3	DW Milk Barn	9/7/23	7:00	621	0.62	1.1	*		7.8
4	DW Millers	9/7/23	7:00	647	0.65	1.1	*		8
5	DW Uncles	9/7/23	7:00	529	0.53	1	*		7.8

* = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott McFrickland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02